Predicting Store-Level Stockouts in Fresh Retail

B.Tech Project Presentation

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2026-08-24

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The Problem Statement

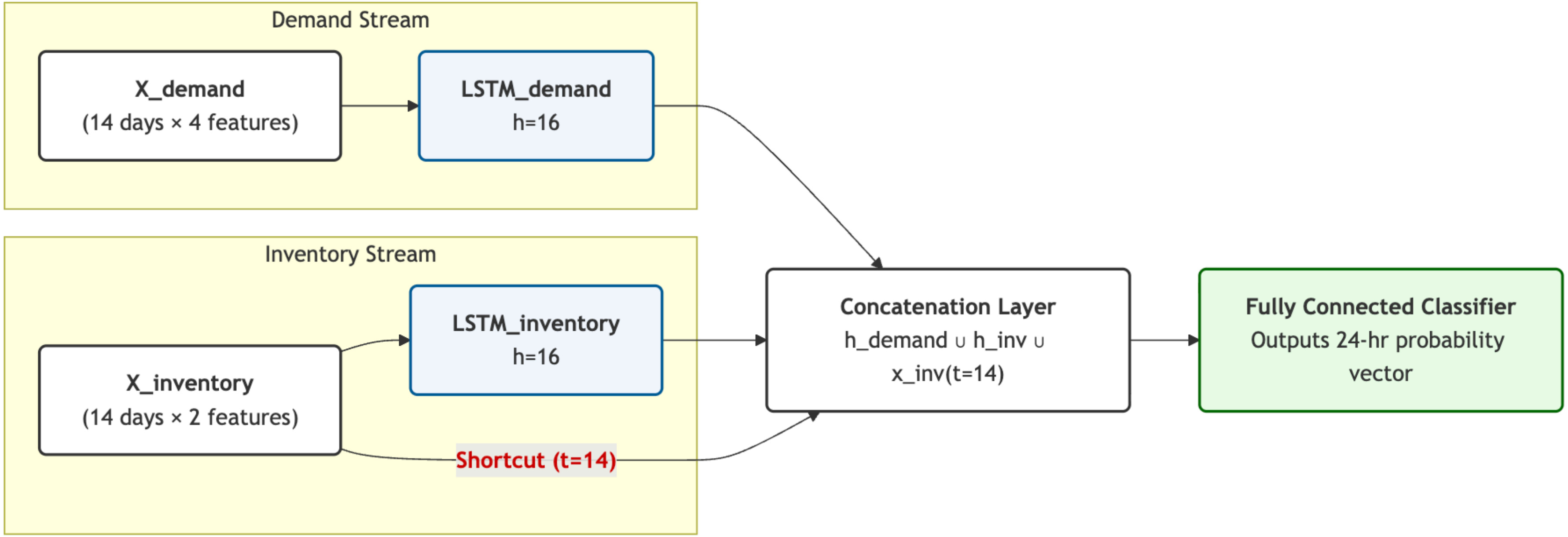
Fresh retail products (produce, meat, dairy) have extremely short shelf lives, creating a tightrope walk for store managers:

- **Overstocking** leads to massive spoilage, financial loss, and food waste.
- **Understocking** leads to stockouts (empty shelves), lost immediate sales, and frustrated customers.

The Challenge: Traditional supply chain forecasting operates at the warehouse level over weekly horizons. However, managing fresh inventory requires store-level, hourly precision.

Our Objective: To build a Deep Learning architecture capable of predicting the exact hourly stockout status of a product within a specific store for the next 24 hours. This allows managers to proactively trigger intra-day replenishment and minimize both waste and lost sales.

Architecture: Dual-Stream Inventory Shortcut



Key Architectural Innovations

To solve the vanishing signal problem and isolate causal factors, we designed a custom LSTM architecture that splits the data and implements a bypass mechanism.

1. Dual-Stream Processing: We separate the 14-day history into two branches:

- **Demand:** Sales, discounts, holidays.
- **Inventory:** Daily stock status.

This prevents erratic daily sales noise from washing out the strict physical reality of stock levels.

2. The Inventory Shortcut: Even LSTMs forget over a 14-day window. The single most important predictor of tomorrow's stockout is **today's** inventory. We physically bypass the recurrent layers and feed today's inventory straight into the final classifier.